An Introduction to Cybersecurity Information Sharing

MISP - Intelligence Sharing

CIRCL / Team MISP Project

MISP Project

https://www.misp-project.org/

CRI MISP 2024

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Agenda

- Agenda and details available
  https://tinyurl.com/CRI-MISP

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MISP and starting from a practical use-case

- During a malware analysis workgroup in 2012, we discovered that we worked on the analysis of the same malware.
- We wanted to share information in an easy and automated way **to avoid duplication of work**.
- Christophe Vandeplas (then working at the CERT for the Belgian MoD) showed us his work on a platform that later became MISP.
- A first version of the MISP Platform was used by the MALWG and **the increasing feedback of users** helped us to build an improved platform.
- MISP is now **a community-driven development**.

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about CIRCL

The Computer Incident Response Center Luxembourg (CIRCL) is a government-driven initiative designed to provide a systematic response facility to computer security threats and incidents. CIRCL is a CERT with national responsibilities for the private sector, communes and non-governmental entities in Luxembourg and is operated by LHC (Luxembourg House of Cybersecurity).

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MISP and CIRCL

- CIRCL is mandated by the Ministry of Economy and acting as the Luxembourg National CERT for private sector.
- CIRCL leads the development of the Open Source MISP threat intelligence platform which is used by many military or intelligence communities, private companies, financial sector, National CERTs and LEAs globally.
- **CIRCL runs multiple large MISP communities performing active daily threat-intelligence sharing**.



Co-financed by the European Union

Connecting Europe Facility

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What is MISP?

- MISP is a **threat information sharing** platform that is free & open source software
- A tool that **collects** information from partners, your analysts, your tools, feeds
- Normalises, **correlates**, **enriches** the data
- Allows teams and communities to **collaborate**
- **Feeds** automated protective tools and analyst tools with the output

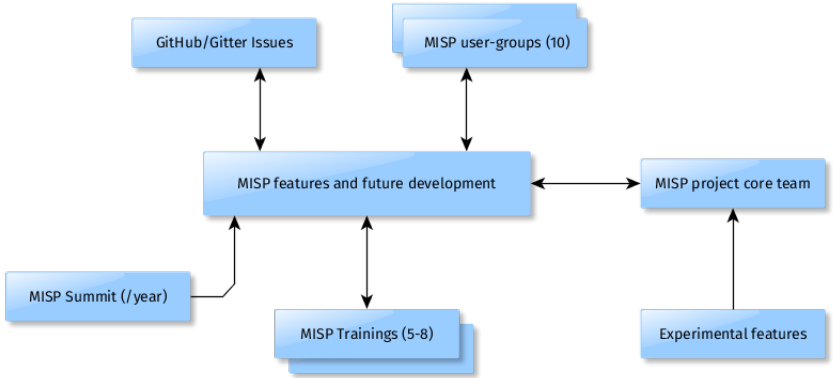
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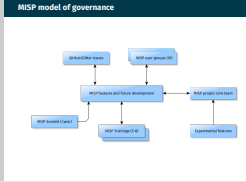
Development based on practical user feedback

- There are many different types of users of an information sharing platform like MISP:
  - **Malware reversers** willing to share indicators of analysis with respective colleagues.
  - **Security analysts** searching, validating and using indicators in operational security.
  - **Intelligence analysts** gathering information about specific adversary groups.
  - **Law-enforcement** relying on indicators to support or bootstrap their DFIR cases.
  - **Risk analysis teams** willing to know about the new threats, likelyhood and occurences.
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MISP model of governance





Many objectives from different user-groups

- Sharing indicators for a **detection** matter.
  - 'Do I have infected systems in my infrastructure or the ones I operate?'
- Sharing indicators to **block**.
  - 'I use these attributes to block, sinkhole or divert traffic.'
- Sharing indicators to **perform intelligence**.
  - 'Gathering information about campaigns and attacks. Are they related? Who is targeting me? Who are the adversaries?'
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Communities using MISP

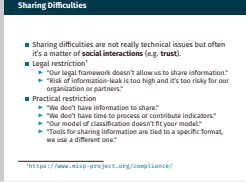
- Communities are groups of users sharing within a set of common objectives/values.
- CIRCL operates multiple MISP instances with a significant user base (more than 1200 organizations with more than 4000 users).
- **Trusted groups** running MISP communities in island mode (air gapped system) or partially connected mode.
- **Financial sector** (banks, ISACs, payment processing organizations) use MISP as a sharing mechanism.
- **Military and international organizations** (NATO, military CSIRTs, n/g CERTs,...).
- **Security vendors** running their own communities (e.g. Fidelis) or interfacing with MISP communities (e.g. OTX).
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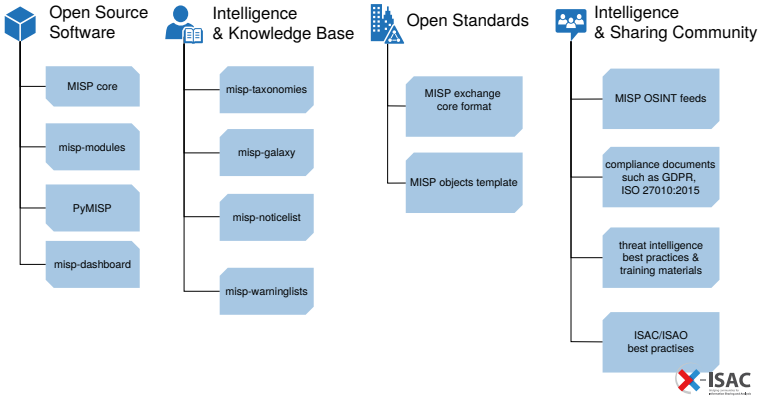
Sharing Difficulties

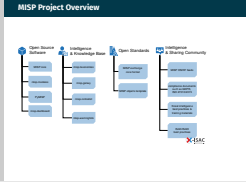
- Sharing difficulties are not really technical issues but often it's a matter of **social interactions** (e.g. **trust**).
- Legal restriction[1]
  - "Our legal framework doesn't allow us to share information."
  - "Risk of information-leak is too high and it's too risky for our organization or partners."
- Practical restriction
  - "We don't have information to share."
  - "We don't have time to process or contribute indicators."
  - "Our model of classification doesn't fit your model."
  - "Tools for sharing information are tied to a specific format, we use a different one."

[1] https://www.misp-project.org/compliance/



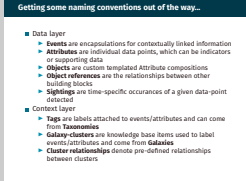
MISP Project Overview





Getting some naming conventions out of the way...

- Data layer
  - **Events** are encapsulations for contextually linked information
  - **Attributes** are individual data points, which can be indicators or supporting data
  - **Objects** are custom templated Attribute compositions
  - **Object references** are the relationships between other building blocks
  - **Sightings** are time-specific occurances of a given data-point detected
- Context layer
  - **Tags** are labels attached to events/attributes and can come from **Taxonomies**
  - **Galaxy-clusters** are knowledge base items used to label events/attributes and come from **Galaxies**
  - **Cluster relationships** denote pre-defined relationships between clusters



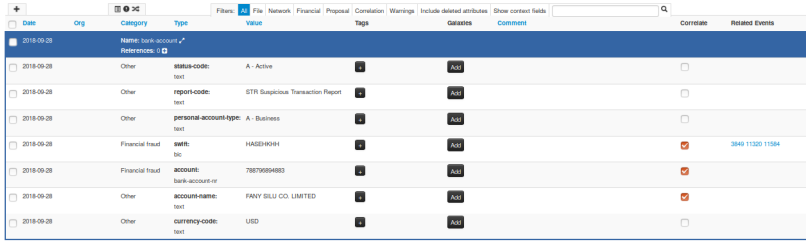
Terminology about Indicators

- Indicators[2]
  - Indicators contain a pattern that can be used to detect suspicious or malicious cyber activity.
- Attributes in MISP can be network indicators (e.g. IP address), system indicators (e.g. a string in memory) or even bank account details.
  - **A type (e.g. MD5, url) is how an attribute is described**.
  - An attribute is always in a category (e.g. Payload delivery) which puts it in a context.
    - **A category is what describes** an attribute.
  - An IDS flag on an attribute allows to determine if **an attribute can be automatically used for detection**.

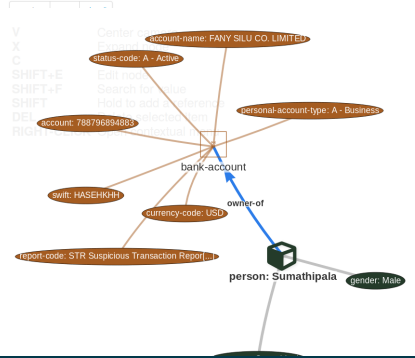
[2] IoC (Indicator of Compromise) is a subset of indicators

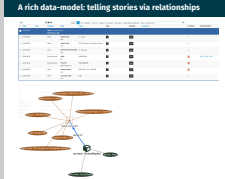
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A rich data-model: telling stories via relationships



Date	Org	Category	Type	Value	Tags	Galaxies	Comment	Correlate	Related Events
2018-09-28		Name: bank-account References: 0							
2018-09-28		Other	status-code: text	A - Active				☐	
2018-09-28		Other	report-code: text	STR Suspicious Transaction Report				☐	
2018-09-28		Other	personal-account-type: text	A - Business				☐	
2018-09-28		Financial fraud	swift: bic	HASEHKHH				☑	3849 11320 11584
2018-09-28		Financial fraud	account: bank-account-nr	78879689883				☑	
2018-09-28		Other	account-name: text	FANY SILU CO. LIMITED				☑	
2018-09-28		Other	currency-code: text	USD				☐	

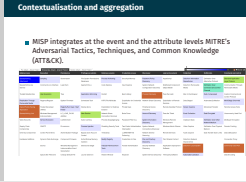




Contextualisation and aggregation

- MISP integrates at the event and the attribute levels MITRE's Adversarial Tactics, Techniques, and Common Knowledge (ATT&CK).





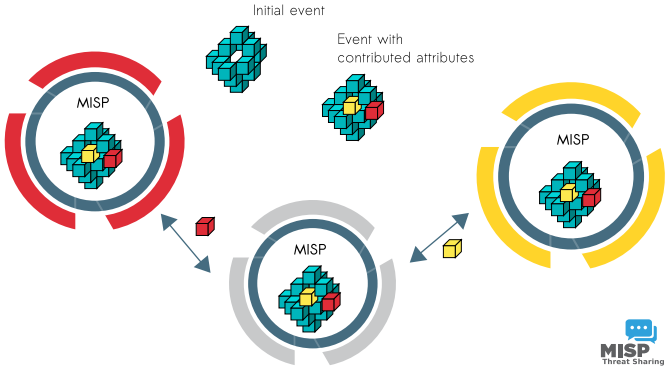
Sharing in MISP

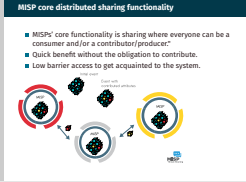
- Sharing via distribution lists - **Sharing groups**
- **Delegation** for pseudo-anonymised information sharing
- **Proposals** and **Extended events** for collaborated information sharing
- Synchronisation, Feed system, air-gapped sharing
- User defined **filtered sharing** for all the above mentioned methods
- Cross-instance information **caching** for quick lookups of large data-sets
- Support for multi-MISP internal enclaves

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MISP core distributed sharing functionality

- MISPs' core functionality is sharing where everyone can be a consumer and/or a contributor/producer."
- Quick benefit without the obligation to contribute.
- Low barrier access to get acquainted to the system.



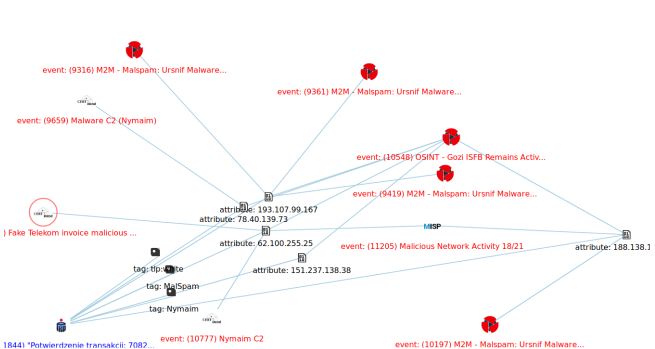


Information quality management

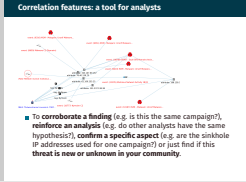
- Correlating data
- Feedback loop from detections via **Sightings**
- **False positive management** via the warninglist system
- **Enrichment system** via MISP-modules
- **Integrations** with a plethora of tools and formats
- Flexible **API** and support **libraries** such as PyMISP to ease integration
- **Timelines** and giving information a temporal context
- Full chain for **indicator life-cycle management**

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Correlation features: a tool for analysts



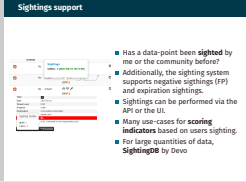
- To **corroborate a finding** (e.g. is this the same campaign?), **reinforce an analysis** (e.g. do other analysts have the same hypothesis?), **confirm a specific aspect** (e.g. are the sinkhole IP addresses used for one campaign?) or just find if this **threat is new or unknown in your community**.



Sightings support

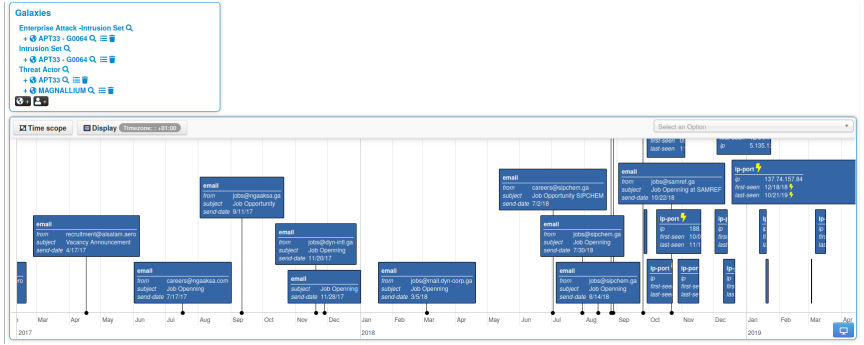


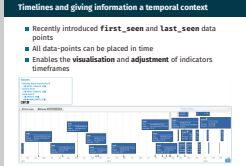
- Has a data-point been **sighted** by me or the community before?
- Additionally, the sighting system supports negative sigthings (FP) and expiration sightings.
- Sightings can be performed via the API or the UI.
- Many use-cases for **scoring indicators** based on users sighting.
- For large quantities of data, **SightingDB** by Devo



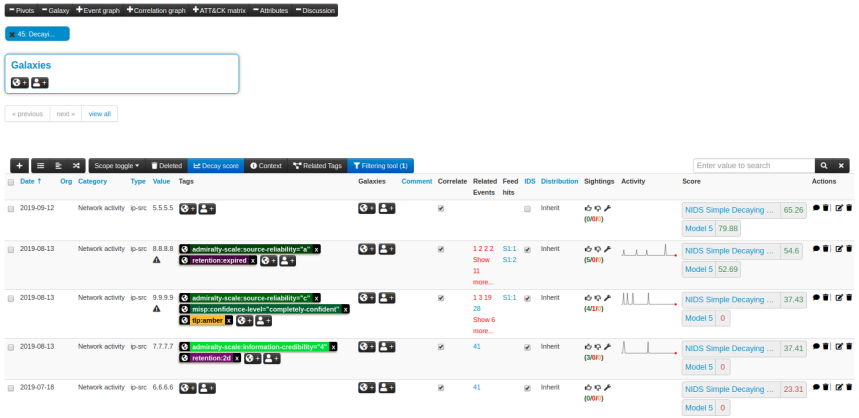
Timelines and giving information a temporal context

- Recently introduced **`first_seen`** and **`last_seen`** data points
- All data-points can be placed in time
- Enables the **visualisation** and **adjustment** of indicators timeframes

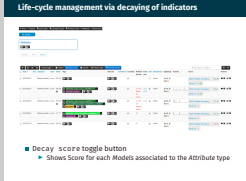




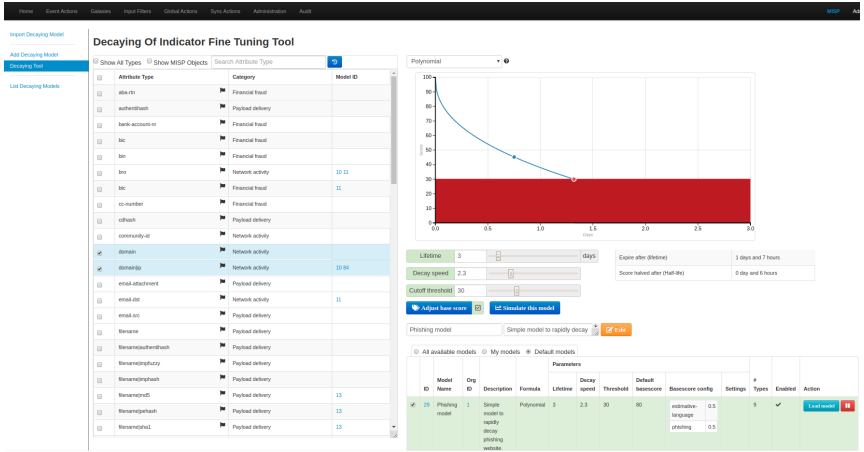
Life-cycle management via decaying of indicators



- `Decay score` toggle button
  - Shows Score for each *Models* associated to the *Attribute* type



Decaying of indicators: Fine tuning tool



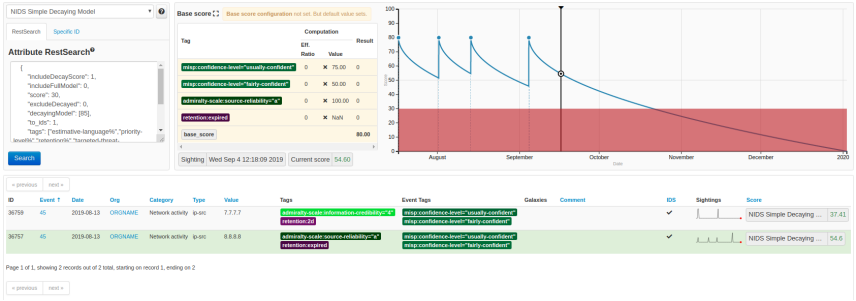
Create, modify, visualise, perform mapping

Decaying of indicators: Fine tuning tool



Create, modify, visualise, perform mapping

Decaying of indicators: simulation tool



Simulate *Attributes* with different *Models*

Decaying of indicators: simulation tool



Simulate *Attributes* with different *Models*

Bootstrapping your MISP with data

- We maintain the default CIRCL OSINT feeds (TLP:WHITE selected from our communities) in MISP to allow users to ease their bootstrapping.
- The format of the OSINT feed is based on standard MISP JSON output pulled from a remote TLS/HTTP server.
- Additional content providers can provide their own MISP feeds. (`https://botvrij.eu/`)
- Allows users to **test their MISP installations and synchronisation with a real dataset**.
- Opening contribution to other threat intel feeds but also allowing the analysis of overlapping data[3].

[3] A recurring challenge in information sharing

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Conclusion

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- Enable users to customize MISP to meet their community's use-cases.
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